Md Ashfaqur Rahman

Dhaka, Bangladesh

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EDUCATION

Khulna University of Engineering & Technology (KUET), Bangladesh Jan 2020 – Sep 2025
Bachelor of Science in Computer Science and Engineering
CGPA: 3.32 / 4.00

RESEARCH INTERESTS

- **Machine Learning**
- **Deep Learning**
- **Computer Vision**
- **Natural Language Processing**
- **Human Computer Interaction**

RESEARCH EXPERIENCE

Undergraduate Thesis Feb 2024 – Aug 2025
Supervisor: **Dr. Muhammad Sheikh Sadi**
Title: *Bridging the Accessibility Gap: A Comprehensive System for Arabic Braille Interpretation*

- Curated and extended an Arabic Braille dataset by collecting samples for underrepresented characters, diacritics, and ligatures, addressing class imbalance in low-resource settings.
- Designed a two-stage recognition pipeline using YOLOv8 for Braille cell detection followed by a custom CNN-based classifier for character recognition.
- Conducted comparative evaluations against existing state-of-the-art approaches, achieving superior accuracy and robustness on complex Braille sequences.
- Developed a language-aware post-processing module to reconstruct syntactically valid Arabic sentences from noisy character-level predictions.
- Performed systematic experimentation including ablation studies, error analysis, and model benchmarking for academic publication.

PUBLICATION

Ashfaqur Rahman and Muhammad Sheikh Sadi, "A Comprehensive System for Arabic Braille Interpretation," *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025, pp. 2992–2997, doi: 10.1109/ICCIT68739.2025.11491613.
[Paper] [Code] [Dataset]

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, PHP, SQL, PL/SQL, GLSL
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, scikit-fuzzy
NLP & LLMs: Hugging Face Transformers, PEFT (LoRA), spaCy, ROUGE/BERTScore evaluation
Computer Vision: OpenCV, Ultralytics YOLOv8
Data & Visualization: NumPy, Pandas, Matplotlib
Software Development: Laravel, Android (MVVM), OpenGL 3.3, Flex/Bison, Pygame
Tools: Git, GitHub Actions, Docker, Linux, LaTeX, Colab, Kaggle

SELECTED PROJECTS

- **Medical Chat SOAP Summarization**
  Fine-tuned Flan-T5-base with LoRA (0.71% of parameters trainable) to convert doctor–patient dialogues into structured SOAP notes, using a 10,000-dialogue dataset. Achieved ROUGE-1 0.503, ROUGE-L 0.369 and BERTScore F1 0.886 on the held-out test set.

- **Mancala Game: AI vs. Player**
  Built a Mancala game in Pygame with an Alpha-Beta pruning agent as the opponent and a genetic algorithm that suggests moves to the human player. A fuzzy-logic module estimates each side's winning probability in real time from the board state.

- **News Brief: A News Summarizer App**
  Built an Android app backed by a Python NLP pipeline that generates extractive news summaries with the TextRank algorithm.

- **Compiler Project**
  Built a compiler front-end for a custom high-level language using Flex and Bison, supporting variables, expressions, control flow, functions and I/O. Implemented semantic analysis, symbol-table management and an interpreter that runs programs end to end.

- **3D Football Stadium**
  Built a real-time 3D stadium simulation in modern OpenGL 3.3 with custom GLSL shaders, Phong lighting from multiple light sources, and texture mapping. Added physics-based ball trajectories and interactive camera controls.

- **CardiacRecorder**
  Built an Android app, using an MVVM architecture, that records, validates and visualizes cardiovascular measurements. Added automated unit and UI tests with continuous integration through GitHub Actions.

- **School Management System**
  Built a full-stack Laravel web application with role-based access control, a normalized relational schema and RESTful CRUD services to automate academic workflows.

- **Banking Database System**
  Designed and implemented a normalized banking database with complex integrity constraints and multi-entity relationships, using SQL and PL/SQL for transactions and queries.

ACTIVITY

- **Event Organizer, BitFest 2025**

  Organized the Inter-University Programming Contest (IUPC) at the 3rd KUET CSE National Festival.

  Skills: Team leadership, event coordination, cross-team communication

LANGUAGES

Bangla: Native
English: Proficient